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- An $(\mathbf{x}^*, \mathbf{u})$ (or $\mathbf{x}^*$) which satisfies the **Karush Kuhn Tucker (KKT)** conditions is called a **KKT** point.

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  Hence $[2, 1]^T$ satisfies the **KKT** condition.

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  5. Does there exist an FJ point which is not a KKT point?

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- If $\mathbf{x}^* \in S$ is such that for all $\mathbf{d} \in D_{\mathbf{x}^*} (\neq \phi)$, $\nabla f(\mathbf{x}^*)\mathbf{d} > 0$ then does it imply that $\mathbf{x}^*$ is a local minimizer of f?

- Consider the following problem:
  Minimize $-x_1^2 - 4x_1x_2 - x_2^2$
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  4. If the objective function is changed to $2x_1^2 - x_1x_2 + x_2^2 - x_2$ and if $S = Fea(P) = \{(x_1, x_2) : x_2^2 + x_1^2 \leq 1\}$ then find all optimum solutions to this problem.

- For a problem **(P)** of the form,

$$\begin{aligned} &\text{Minimize } f(\mathbf{x}) \\ &\text{subject to } g_i(\mathbf{x}) \leq 0, \text{ for } i = 1, \ldots, m, \quad \mathbf{x} \in \mathbb{R}^n, \end{aligned}$$

  where all the g_i's and f are continuously differentiable throughout $\mathbb{R}^n$, and S is the feasible region of **(P)** check the correctness of the following statements with proper justification.

  1. If $\mathbf{x}^* \in S$ is a KKT point of the above problem then $-\nabla f(\mathbf{x}^*)$ lies in the cone generated by $\nabla g_i(\mathbf{x}^*), i \in I$, where I gives the indices of the binding constraints (given by g_i's) at $\mathbf{x}^*$.

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  4. If $\mathbf{x}^*$ is an FJ point and there is a solution to the FJ conditions at $\mathbf{x}^*$ with $u_0 = 0$ (u_0 is the coefficient of $\nabla f(\mathbf{x}^*)$ in the FJ conditions), then $\mathbf{x}^*$ is not a KKT point.

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- If $\mathbf{x}^*$ is an FJ point and $\{\nabla g_i(\mathbf{x}^*)\}_{i \in I}$ is **LD**, then $G_{0,\mathbf{x}^*} = \phi$ (or in other words there exists a solution to the FJ conditions with $u_0 = 0$).

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- If $\mathbf{x}^*$ is a local minima of (P) with $F_{0,\mathbf{x}^*} \neq \phi$ but $G_{0,\mathbf{x}^*} = \phi$ then $\mathbf{x}^*$ is not a KKT point.

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- Give examples of nonconstant functions on $\mathbb{R}^n$ which are both convex and concave and those which are neither convex nor concave.

- For a linear programming problem (P) of the form,
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- Consider the linear programming problem.
  Minimize $2x_1 - 3x_2$
  subject to $x_1 + 2x_2 \leq 3$
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- Check whether $[1, 0]^T$ is a KKT point of the above problem. How many feasible points does this problem have? What is your conclusion?